Constraints:

- from ex.3 → polling period of the line sensor reading:

$$t = 100\,mS \longrightarrow f = \frac{1}{t} = \frac{1}{10^{-1}}\,s^{-1} = 10\,Hz$$

Parameters choice:

Since $\text{INTERRUPT FREQUENCY} = \dfrac{\text{TIMER CLOCK FREQUENCY}}{(PSC+1)\times(ARR+1)}$

and $\text{TIMER CLOCK FREQUENCY} = 96\,MHz$

$$\Longrightarrow \quad 10 = \frac{96 \cdot 10^6}{(PSC+1) \times (ARR+1)}$$

$$(PSC+1) \times (ARR+1) = 96 \cdot 10^5$$

$\Longrightarrow$

$PSC = 959$

$ARR = 9999$

$\longrightarrow \underbrace{(959+1)}_{96 \cdot 10} \underbrace{(9999+1)}_{10^4} = 96 \cdot 10^5$

Parameters choice based on simplicity and readability.